

Employee Attrition Prediction using Machine Learning - Project Summary

Project Overview

In this project, I analyzed employee data to understand why employees leave the company and to identify the factors that affect attrition. The aim was to find useful patterns that can help the HR team improve employee retention and reduce employee turnover.

Key Findings

From the analysis, I found that about 16% of employees left the company. The Sales department, especially Sales Representatives, had the highest attrition rate. Employees who worked overtime, had lower monthly income, and had spent fewer years at the company were more likely to leave. I also observed that salary is not the only reason employees leave. Work-life balance, job satisfaction, and overtime also have a strong impact on employee attrition.

Recommendations

Based on these findings, I think HR should focus more on employees during their first few years in the company because this is when most employees tend to leave. HR can also arrange regular feedback sessions with employees who work overtime or seem dissatisfied. Improving work-life balance, providing career growth opportunities, and recognizing employee efforts can help increase job satisfaction and reduce attrition. Since the Sales department has the highest attrition, it should be given priority for retention programs.

Limitation

This analysis is based only on the available employee data. It cannot identify personal reasons for leaving, such as family responsibilities, career changes, or better job opportunities outside the company. Therefore, these results should be used to support HR decisions along with employee feedback.

Conclusion

Overall, this project helped identify the main factors related to employee attrition and showed how data can help HR make better decisions. By focusing on employees who are more likely to leave and improving workplace policies, the company can improve employee satisfaction and reduce future attrition.